CO1-AT1

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 06\1 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks: 25**

Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct:

I M.Ajay/192312542 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M.Ajay.

Annexure A

Short Answer Test

1. A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C? (5 Marks)

2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C? (5 Marks)

3. In a real-world medical image classification task for detecting breast cancer, a deep-learning model integrated with the K-Nearest Neighbors algorithm is applied for binary classification (malignant vs. benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True Positives (TP), 190 True Negatives (TN), and 30 False Positives (FP). Based on the given information, determine the missing False Negatives (FN) value, and then calculate the Accuracy, Precision, Sensitivity (Recall), and Specificity of the classifier, explaining its effectiveness in medical diagnosis. (5 Marks)

4. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)

(a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
(b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
(c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
(d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
(e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
(f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
(g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
(h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

i) Identify the epoch at which the model begins to overfit.
ii) Suggest two effective techniques to reduce overfitting in this model.
iii) Discuss how overfitting affects the model's generalization ability on unseen data.

5. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)

(a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%
(b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%
(c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%
(d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%
(e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%
(f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%
(g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%
(h)Epoch 8: Training Accuracy = 92%,Validation Accuracy = 68%
i) Identify the epoch at which the model starts overfitting.
ii) Explain why validation accuracy decreases despite training accuracy increasing.
iii) Suggest two methods to improve validation performance.

6. A machine learning model shows the following loss values during training: (5 Marks)

(a) Epoch 1: Training Loss = 3.0, Validation Loss = 2.7
(b) Epoch 2: Training Loss = 2.6, Validation Loss = 2.3
(c) Epoch 3: Training Loss = 2.3, Validation Loss = 2.1
(d) Epoch 4: Training Loss = 2.1, Validation Loss = 2.0
(e) Epoch 5: Training Loss = 1.9, Validation Loss = 2.2
(f) Epoch 6: Training Loss = 1.7, Validation Loss = 2.5
(g) Epoch 7: Training Loss = 1.6, Validation Loss = 2.8
(h) Epoch 8: Training Loss = 1.4, Validation Loss = 3.1
i) At which epoch does the model achieve optimal generalization?
ii) Identify signs of overfitting from the data.
iii) Recommend two regularization techniques and explain their impact.

7. In an industrial process, an AI robot is assigned to segregate the front and back tires automatically. Let the number of tires are 250 (125 front and 125 back tires). (5 Marks)

a) Front identified as Front: 119
b) Front identifies as Back: 06
c) Back identified as Back: 120
d) Back identified as Front: 05

Compute: Accuracy, Precision, Sensitivity, Specificity and F1-Score.

8. In a medical diagnosis system, an AI model detects whether a patient has a disease or not. Out of 300 cases (150 diseased and 150 healthy): (5 Marks)

a) Diseased identified as Diseased: 138
b) Diseased identified as Healthy: 12
c) Healthy identified as Healthy: 135
d) Healthy identified as Diseased: 15

Compute: **Accuracy, Precision, Sensitivity (Recall), Specificity, and F1-Score.**

Subname: Machine learning
Course code: ITA611

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COI-AII

(23) Given

$P(A)=\frac{1}{4}$, $P(B)=\frac{1}{2}$, $P(C)=\frac{1}{4}$

chance of drawing a blue ball from each box

$P(blue|A)=\frac{4}{12}=\frac{1}{3}$, $P(B|ue(B)=\frac{7}{15}$, $P(blue|C)=\frac{2}{15}$

Find the overall probability of drawing blue

$P(blue)=\frac{1}{4}\cdot\frac{1}{3}+\frac{1}{2}\cdot\frac{7}{15}+\frac{1}{4}\cdot\frac{2}{15}=\frac{1}{12}+\frac{7}{30}+\frac{1}{30}=\frac{23}{60}$

$P(C|blue)=\frac{P(blue|C\ P(C))}{P(blue)}=\frac{\frac{2}{15}\cdot\frac{1}{4}}{\frac{23}{60}}=\frac{2}{23}=8.70\%$

(24) Given data

$P(A)=\frac{1}{4}$, $P(B)=\frac{1}{2}$, $P(C)=\frac{1}{4}$

chance of selecting a blue car from each section

$P(blue|A)=\frac{3}{10}$, $P(blue|B)=\frac{6}{15}=\frac{2}{5}$, $P(blue|C)=\frac{2}{15}$

overall probability of selecting a blue car

$P(blue)=\frac{1}{4}\cdot\frac{3}{10}+\frac{1}{2}\cdot\frac{2}{5}+\frac{1}{4}\cdot\frac{2}{15}=\frac{3}{40}+\frac{1}{5}+\frac{1}{30}=\frac{27}{120}$

$$P(C/blue) = \frac{P(blue/C)P(C)}{P(blue)} = \frac{\frac{2}{15} \cdot \frac{1}{4}}{\frac{27}{120}} = \frac{4}{27} = 14.81\%$$

(34) Given

There are 1800 samples in each of two classes

$$1800 + 1800 = 3600$$

Size of the test set

$$= 3600 \times 20\%$$

$$= 720 \text{ test samples}$$

The missing false negatives

$$FN = 720 - (210 + (90 + 30)) = 290$$

Each metric

Accuracy: proportion of all predictions that are correct

$$\frac{TP + TN}{TP + TN + FP + FN} = \frac{210 + 190}{720} = 55.56\%$$

Precision: Proportion of predicted malignant cases that are actually malignant

$$\frac{TP}{TP + FP} = \frac{210}{210 + 30} = 87.50\%$$

Sensitivity: Proportion of Actual benign cases the

$$\frac{TP}{TP + FN} = \frac{210}{210 + 290} = 42.0\%$$

Specificity: proportion of Actual benign cases the model correctly identifies

$$\frac{TN}{TN+FP} = \frac{190}{190+30} = 86.36$$

Accuracy

$$= \frac{TP+TN}{total} \times 100$$

$$= \frac{210+190}{720} \times 100$$

$$= 55.6\%$$

(4) i) overfitting starts at each 4

→ sup to EPOCH 3, both training and validation loss decrease

→ From EPOCH 4, training loss decreases (2.0-3) but validation loss increases (2-1.5 3.2)

→ So the model starts learning the training data to much

ii) Two techniques to reduce overfitting:

* Dropout - randomly removes some neutrons during training

* Early stopping stops training when validation loss starts increasing

iii) Effect on generalization:

Overfitting makes the model perform well on training data but poorly on unseen new data.

5) (i) Overfitting starts

Overfitting starts at Epoch 5 because validation accuracy reaches its highest value of 73% and then and then decreases while training accuracy continues increasing

(ii) Reason:

The model memorizes the training instead of learning but Accuracy on unseen validation data decreases.

(iii) Two methods to improve validation Performance

* Dropout Regularization - reduces overfitting
* Early stopping training when validation Accuracy stops improving.

Conclusion:

The best model is around Epoch 15 with 80% training Accuracy and 73% validation Accuracy.